

Submit the following five exercises. You are expected to provide full solutions in a linear and coherent fashion. You are encouraged to work in groups, but your work must be your own and must be self-contained.

Exercise 1.

Consider the function $f(x) = -\frac{1}{2}x + 5$.

1. Approximate the area under the curve on the interval $[0, 4]$ using Riemann sums, with left endpoints, and two bars.
2. Approximate the area under the curve on the interval $[0, 4]$ using Riemann sums, with left endpoints, and four bars.
3. Compute the exact area by drawing a picture and using formulas for areas.
4. Compute the exact area by integrating.
5. Explain which of the approximations is better. Explain if the approximations are overestimates or underestimates. Explain why you would have expected this at the start of the problem.

Exercise 2.

Consider the function

$$g(x) = \begin{cases} 2(x+1) + 6 & \text{if } x \leq 1, \\ -x + 7 & \text{if } 1 < x. \end{cases}$$

1. Sketch the graph of $g(x)$.
2. Compute $\int_{-1}^3 g(x)dx$.

Exercise 3.

1. Compute $\int_0^\pi (\sin(x) + x^3 + 2)dx$.
2. Compute $\int 2dx$.
3. Compute $\int \cos(3x)dx$.
4. Compute $\int e^{x+1}dx$.

Exercise 4.

Let $f(x)$ and $g(x)$ be functions with $\int_{-1}^3 f(x)dx = 6$ and $\int_{-1}^3 g(x)dx = -3$, respectively.

1. Compute $\int_{-1}^3 (2f(x) - g(x))dx$.
2. Compute $\int_{-1}^3 (4f(x) + 5g(x))dx$.

Exercise 5.

Compute $\int \frac{\sqrt{x}-2x^4}{\sqrt{x^3}}dx$.